Making Tail Constraints Bind: Two Silent Failure Modes in CVaR-Constrained Portfolio RL

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ACM paper
GitHub: nl2992/cvar-rl-portfolio-allocator

Abstract

Reinforcement learning (RL) portfolio allocators are often given a tail-risk limit through a Lagrangian penalty on Conditional Value-at-Risk (CVaR). We show two ways in which this limit fails with no visible symptom, and fix them by scaling the dual step to the CVaR excess and adding a decision-time filter that moves each over-budget portfolio toward minimum variance before it trades.

7.4% out-of-sample breach rate with the filter, down from 75.1%

26 of 27 untied walk-forward folds with lower $\mathrm{CVaR}_{0.99}$, $p = 8.8 \times 10^{-6}$

7.7% drawdown in March 2020, against 16.3% without the constraint

Motivation

In mid-March 2020 the S&P 500 lost about 15% in a single week. An RL allocator trained only to maximise return stayed in equities and fell 16.3% from its peak. The same allocator with a limit on its tail losses had already moved into Treasuries and cash and fell 7.7% (Fig. 1). One crash is a single case, so Figure 8 repeats the comparison over calm, high-volatility and selloff weeks, where the limit also cut tail losses.

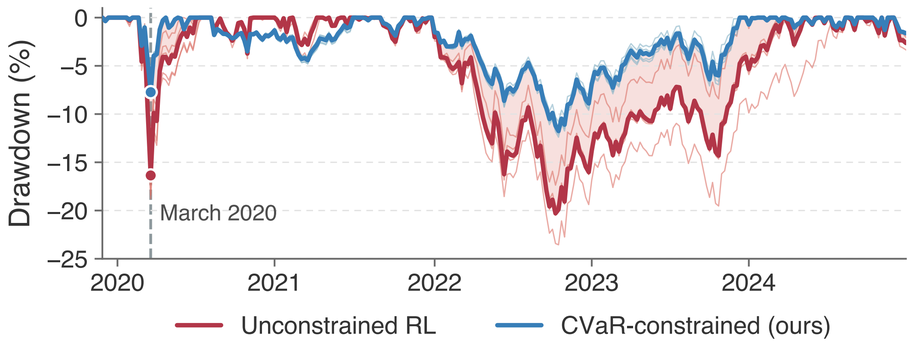


Figure 1. Drawdown over 2020 to 2024. Thick lines are five-seed means and thin lines individual seeds. Shaded, the drawdown the constraint avoided.

Problem and measures

CVaR_α is the average loss in the worst $(1-\alpha)$ share of weeks [1]. With $\alpha = 0.95$ it averages the worst 5%, so it measures how large the losses are beyond the Value-at-Risk (VaR) threshold (Fig. 2).

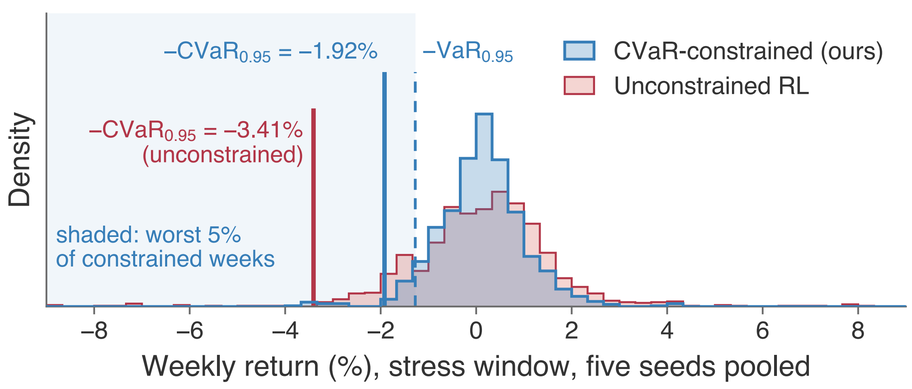


Figure 2. Weekly returns on the stress window, five seeds pooled. Shaded, the worst 5% of constrained weeks. Lines sit at minus each CVaR.

Each week the allocator chooses long-only weights w_t on seven ETFs (SPY, TLT, HYG, DBC, GLD, UUP, BIL), using 887 weekly returns from 2008 to 2024. The problem is a constrained Markov decision process [2],

$$\max_{\pi} J(\pi) \quad \text{s.t.} \quad \mathrm{CVaR}_{0.95}(L) \le d = 1.2\%, \qquad \mathrm{CVaR}_\alpha(L) = \min_{\zeta \in \mathbb{R}} \Big\{ \zeta + \frac{1}{1-\alpha} \mathbb{E}\big[(L-\zeta)^+\big] \Big\}.$$

A test week is a breach when the trailing 52-week $\mathrm{CVaR}_{0.95}$ exceeds d, so the breach rate over the test weeks $\mathcal{T}$ is

$$B = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \mathbf{1}\big[\widehat{\mathrm{CVaR}}_{0.95}(r_{t-52:t-1}) > d\big],$$

with 5% of weeks tolerated. Sharpe ratios are annualised from weekly net returns, and their significance is judged with the Probabilistic and Deflated Sharpe ratios [4].

Failure 1. The penalty weight never moves

With a dual step $\eta_\lambda = 10^{-3}$, λ stayed near zero for all 1500 training updates (Fig. 3). The CVaR excess is about 10^{-2} per week, so each update moved λ by about 10^{-5}. Starting from $\lambda_0 = 0$, after K updates

$$\lambda_K \;\le\; \eta_\lambda \sum_{k=1}^{K} \big(\widehat{\mathrm{CVaR}}_k - d\big)^+ \;\approx\; 10^{-3} \cdot 1500 \cdot 10^{-2} \;\approx\; 10^{-2},$$

and the trained value was 0.022. With $\eta_\lambda = 5.0$, λ rose to between 0.1 and 0.3. A model-free actor-critic failed in the same way when its cost signal (about 10^{-3}) was left unstandardised next to a unit-scale reward signal.

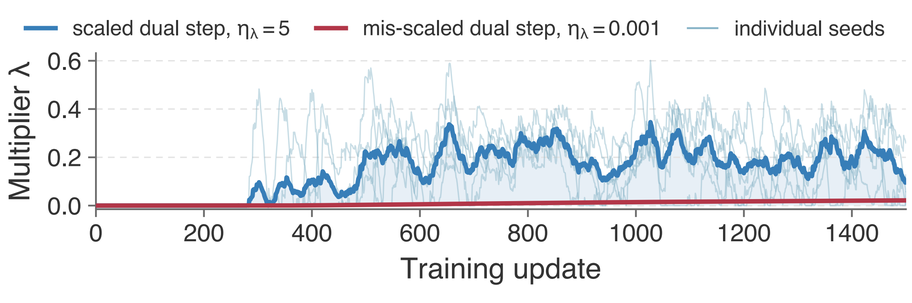


Figure 3. The penalty weight λ during training, five seeds.

Tail metrics do not reveal the failure. Even with λ near zero, the rest of the pipeline (the anchor, the projection and checkpoint selection under the CVaR limit) lowered $\mathrm{CVaR}_{0.99}$ by 61% against the unconstrained learner (0.025 against 0.065, permutation test $p = 0.0044$).

Remedy. The multiplier should be monitored during training, since a value that remains at zero indicates an inactive constraint, and the dual step should be set on the scale of the CVaR excess. With $\eta_\lambda = 5.0$ the penalty binds in training, although the portfolio still exceeds its budget out of sample. The decision-time filter addresses this residual breach.

Failure 2. The test protocol reverses the ranking

A walk-forward test retrains the model on a rolling window and evaluates it on the next unseen block. On a 31-asset universe, one fixed stress split ranked the constrained learner above minimum variance (Sharpe 0.85 against 0.23). A 20-fold walk-forward reversed the order (0.74 against 1.40, Fig. 4). With Ledoit-Wolf shrinkage [5], minimum variance still reached only 0.25 on the split. The tail-risk reduction against unconstrained RL held under both tests (fold-level Wilcoxon $p = 6 \times 10^{-4}$).

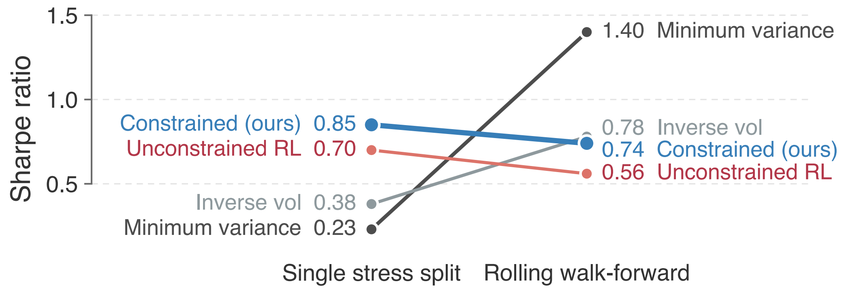


Figure 4. The same four methods ranked by two tests, 31-asset universe.

Remedy. Learned allocators should be compared with classical optimisers on rolling walk-forward folds, with multiple-testing control and the Deflated Sharpe ratio [4]. The Results section reports both protocols.

Our method

We train the allocator end to end with a CVaR penalty whose dual step matches the scale of the CVaR excess, and a filter checks every portfolio before it trades. Figure 5 shows the weekly decision as four maps, defined in steps (i) to (iv).

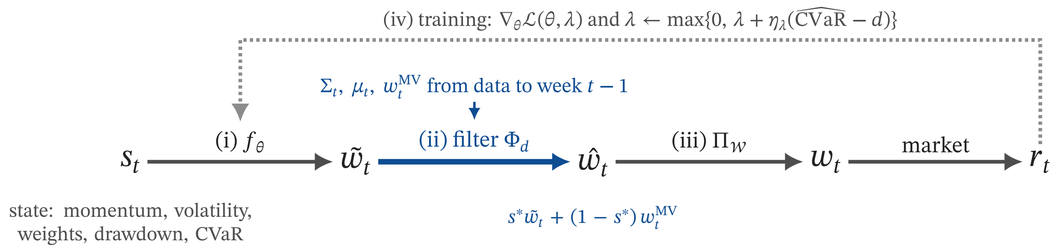


Figure 5. The decision at week t as a composition of maps.

(i) *Policy.* An MLP f_θ tilts an adaptive inverse-volatility anchor a_t, $\tilde{w}_t = \mathrm{softmax}(\log a_t + f_\theta(s_t))$. At initialisation $f_\theta \approx 0$, so training starts from the anchor and learns a state-dependent tilt.

(ii) *Filter.* From the trailing 104 weekly returns it forecasts $\widehat{\mathrm{CVaR}}(w) = c_\alpha \sigma_p(w) - \mu_p(w)$, with $c_\alpha = \varphi(\Phi^{-1}(\alpha))/(1-\alpha)$ and $c_{0.95} = 2.06$. An over-budget proposal is mixed with the trailing minimum-variance portfolio w_t^{MV}, keeping as much of the proposal as the budget allows, $s^* = \max\{s \in [0,1] : \widehat{\mathrm{CVaR}}(s\tilde{w}_t + (1-s)w_t^{\mathrm{MV}}) \le d\}$. It is a safety layer in the sense of [3] and uses data up to week $t-1$ only.

(iii) *Projection.* Π_W enforces long-only weights, $w_i \le 0.40$ and $\|w_t - w_{t-1}\|_1 \le 0.50$. The market then pays r_t net of a 5 bps cost per unit of turnover.

(iv) *Training.* On random windows of past weeks we back-propagate

$$\mathcal{L}(\theta, \lambda) = -\mathrm{Sharpe}(r) + \lambda\, \mathrm{ReLU}\big(\widehat{\mathrm{CVaR}} - d\big) + \kappa\, \mathrm{turnover}, \qquad \lambda \leftarrow \max\{0,\ \lambda + \eta_\lambda(\widehat{\mathrm{CVaR}} - d)\},$$

where $\widehat{\mathrm{CVaR}}$ is the mean of the worst 5% of window returns, so its gradient reaches only tail weeks. We keep the checkpoint with the best validation objective among those within the validation CVaR limit.

Setup. The stress test trains on 2008 to 2018 and tests on the 267 weeks from late 2019 to 2024, with seeds 7, 13, 23, 42 and 2025. Six classical rules (equal weight, inverse volatility, minimum variance, mean-variance, risk parity and a min-CVaR LP) are refit weekly on the trailing 104 weeks.

Results

Decision-time filter. A larger dual step made λ move, but the portfolio was then over budget in 75.1% of test weeks. Classical rules met the same budget in 92% of weeks, so the budget was attainable. Adding the filter lowered the breach rate to **7.4%** (Fig. 6). Raising λ during testing without the filter left the breach rate at 81.9%, which attributes the improvement to the filter.

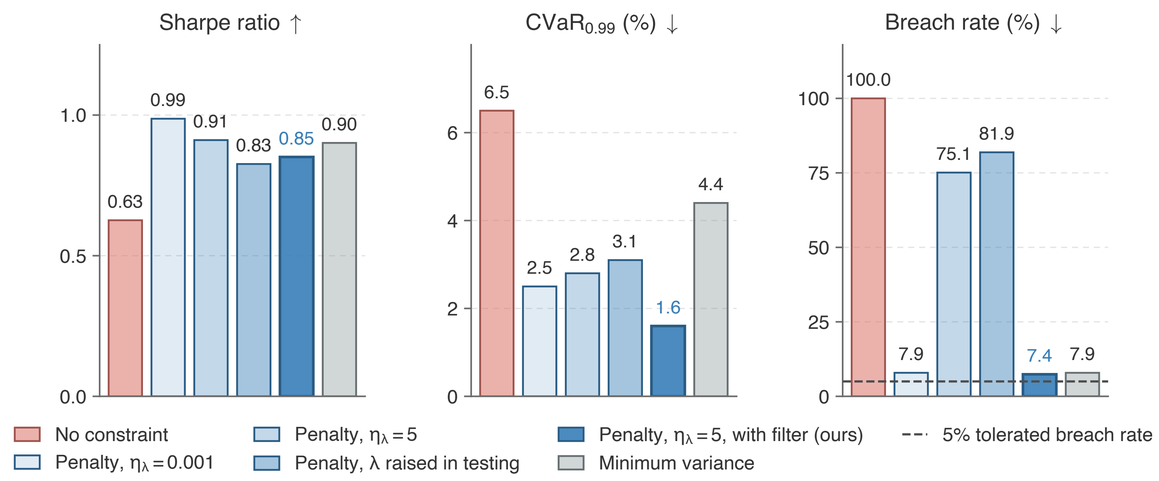


Figure 6. Stress window, five-seed means. Breach is the share of test weeks over budget.

Tail risk across test settings. The walk-forward test retrains every 26 weeks on the previous six years and tests on the next 26 unseen weeks, giving 20 folds per universe from 2015 to 2024. In Figure 7 each fold is one point. Below the diagonal the constraint lowered $\mathrm{CVaR}_{0.99}$. On it the budget never bound, so both models held the same portfolio.

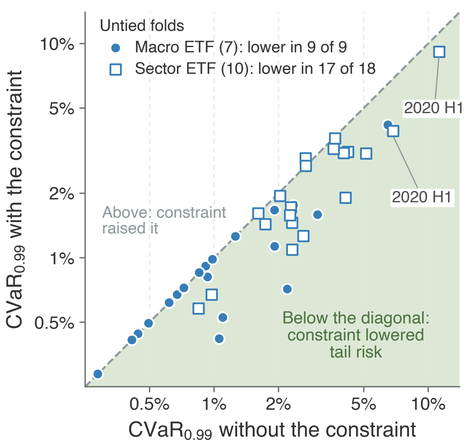


Figure 7. Walk-forward folds, $\mathrm{CVaR}_{0.99}$ without against with the constraint (log scale).

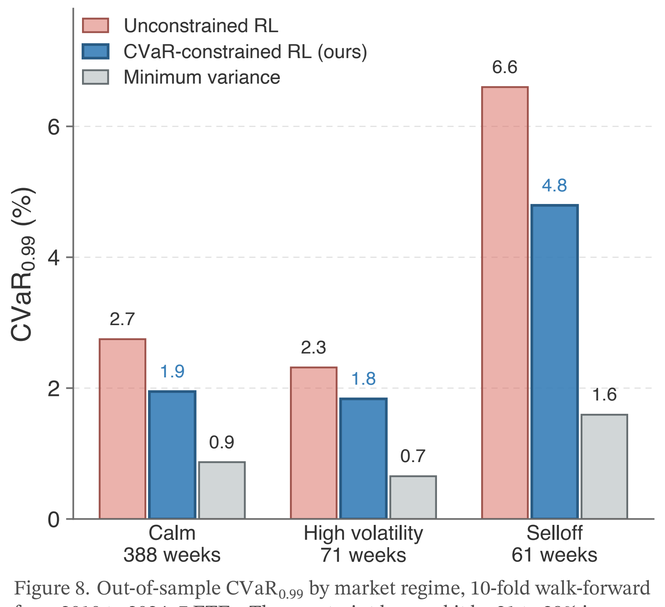


Figure 8. Out-of-sample $\mathrm{CVaR}_{0.99}$ by market regime, 10-fold walk-forward from 2010 to 2024, 7 ETFs. The constraint lowered it by 21 to 29% in every regime.

Table 1. $\mathrm{CVaR}_{0.99}$ (%) without (unc.) and with (con.) the constraint.

Stress window, 267 test weeks

Universe	seeds	unc.	con.	95% CI (pp)
7 ETFs	5	6.47	3.27	[−3.5, −2.2]
10 sector ETFs	3	10.9	7.6	[−5.0, −3.2]

Walk-forward, 20 folds per universe

Universe	unc.	con.	lower	p
7 ETFs	1.43	1.02	9/9	0.0077
10 sector ETFs	3.36	2.50	17/18	0.0003
Both, pooled	**2.40**	**1.76**	**26/27**	8.8×10^{-6}

CI: 95% bootstrap, seed 7. Lower: folds improved, ties excluded. p: Wilcoxon signed-rank.

Limitations and future work

Limitations. The evidence is limited to two equity-centric ETF universes at weekly frequency, under proportional transaction costs without market impact or borrowing costs. The filter relies on a Gaussian approximation to forward CVaR, which may understate losses when returns are fat-tailed, and it achieves compliance partly by holding portfolios close to minimum variance (mean L_1 distance 0.48 to 0.61). The constrained learner underperformed minimum variance in selloff weeks (Sharpe −0.809 against 0.035), its out-of-sample Sharpe advantage over classical baselines did not survive deflation, and the protocol reversal is documented on a single 31-asset window.

Future work. Replacing the Gaussian forecast with a semi-parametric or extreme-value estimate of tail risk would test whether breach depth can be reduced further. An adaptive dual update, such as the PID Lagrangian of [7], could remove the need to tune η_λ against the scale of the CVaR excess. Evaluation on daily data with market-impact costs, on non-equity universes and over multiple rolling start dates would establish how generally the breach reduction and the protocol reversal hold.

References. [1] R. T. Rockafellar, S. Uryasev. Optimization of conditional value-at-risk. *J. Risk* 2 (2000). [2] E. Altman. *Constrained Markov Decision Processes*. 1999. [3] G. Dalal et al. Safe exploration in continuous action spaces. arXiv:1801.08757, 2018. [4] D. H. Bailey, M. López de Prado. The deflated Sharpe ratio. *JPM* 40 (2014). [5] O. Ledoit, M. Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *JMVA* 88 (2004). [6] Y. Benjamini, Y. Hochberg. Controlling the false discovery rate. *JRSS B* 57 (1995). [7] A. Stooke, J. Achiam, P. Abbeel. Responsive safety in RL by PID Lagrangian methods. ICML 2020.

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